

Automated discovery of nonlocal photonic gates

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Interactions between quantum systems enable quantum gates, the building blocks of quantum information processing. In photonics, direct photon-photon interactions are too weak to be practically useful, so effective interactions are engineered using linear optics and measurement. A central challenge is to realize such interactions nonlocally, i.e., between photons that remain spatially separated. We present experimental proposals for several essential nonlocal multiphoton quantum gates acting on spatially separated photons, in both qubit and high-dimensional qudit systems. Rather than relying on preshared entanglement or Bell-state measurements, our gates use quantum indistinguishability via path identity as a resource, exploiting coherent superpositions of photon-pair origins. The mechanisms of the solutions share many properties with quantum teleportation, but do not require shared entanglement or Bell-state measurements. The underlying idea was generated by the multiagent large language model (LLM) system AI-MANDEL, which autonomously implemented its proposed idea using the AI-driven discovery system PyTheus. The human authors then interpreted, generalized, and contextualized the resulting solutions. Technically, our results establish path indistinguishability as a practical resource for distributed quantum information processing; conceptually, they demonstrate how automated discovery systems can contribute ideas and techniques in physics.

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Introduction. Photons are widely regarded as ideal candidates for quantum information carriers due to their negligible decoherence and the relative ease of implementing single-qubit operations. Their weak interaction with the environment allows quantum states to remain coherent over long distances, both in free space and in optical-fiber-based quantum networks. However, this feature comes with a downside—it is extremely challenging to perform multiphoton gates.

Direct photon-photon interactions in free space are basically negligible [1], and their interactions in materials require exceptionally high nonlinearities [2] or intricate mediation via precisely controlled atoms [3]. Fortunately, effective photon-photon interactions can be realized through linear optics

combined with heralding or postselection based on specific detector click patterns [4–8], enabling multiphoton gates essential for linear optical quantum information processing [9] and demonstrated in small photonic processors executing multigate circuits [10]. These gates usually require photons to either meet at one location (e.g., a beam splitter) or share entangled ancilla states or Bell-state measurements.

Recently, a different physical principle—path identity and the indistinguishability of photon origins—has emerged as an alternative resource for photonic quantum information processing [11–13]. For example, various state generation protocols have been proposed that can generate complex quantum states [14,15], even nonlocally [16]. Moreover, a series of recent experiments have demonstrated the feasibility of path identity for state generation [17–20] and observed the underlying multiphoton interference effects for the first time [21–23].

Here, we show that indistinguishability by path identity can also be used to implement nonlocal photonic gates—without the photons ever interacting, sharing a common location, or requiring the need to share entangled ancillary photons. All of the solutions are *feedforwardable* [4], meaning that they could be used as part of quantum circuit and do not need any measurements or conditioning directly after the gate (rather, only after the circuit at each of the nonlocal locations). This

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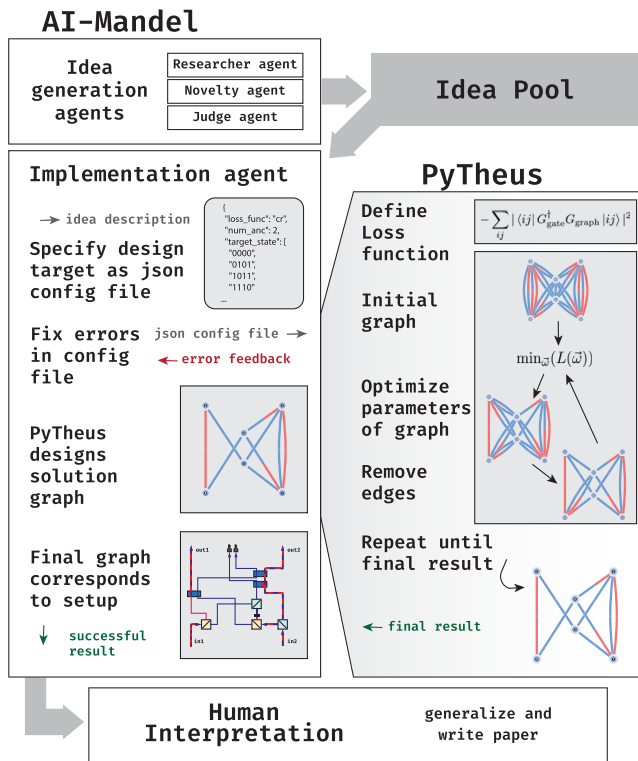


FIG. 1. Idea generation with AI-MANDEL [24]: The core idea of this article has been discovered by a multi-LLM-agentic system AI-MANDEL that has access to an intelligent tool—PyTheus. A set of LLM agents (with domain specific information) generate idea and evaluate the novelty and feasibility of the idea. If they agree, then the idea is stored—in english language—in an idea pool. An implementation agent then takes the idea and translates it into a JSON file with specific instructions for PyTheus to run. PyTheus returns either the final solution (in case of success) or detailed error messages. We have then interpreted, generalized, and conceptualized the solutions. The present work presents one of the most interesting solutions discovered by AI-MANDEL so far.

makes them practically applicable for operations among distant quantum devices in large quantum networks.

The proposals show that path identity and the indistinguishability of the origins of photons can be seen as a resource for photonic quantum information processing. The ideas demonstrated here were generated using the LLM Agent system AI-MANDEL [24] and implemented by the automated design algorithm PyTheus [27]—demonstrating how artificial intelligence (AI) can help discovering principles in physics.

Idea generation via AI-MANDEL—Because the origin of the ideas presented here is unusual, we provide a more detailed description of the underlying AI workflow in Fig. 1. The original research direction did not come from a human, but from an LLM-based multiagent system called AI-MANDEL [24]. The goal of AI-MANDEL is to generate interesting research ideas in quantum physics and implement them with a domain-specific AI system called PyTheus. Each LLM agent in AI-MANDEL has its own memory (context) and is executed via calls to OpenAI’s o4-mini.

AI-MANDEL consists of two main components. The first component aims to generate interesting research ideas in quantum optics in natural language and store them in an idea pool. There, three LLM agents interact. The researcher agent generates initial seeds of research ideas using, as input in its prompt, papers from the arXiv (which it can access autonomously), impact information about previous research [28,29], domain-specific information about PyTheus, and a description of its task, namely, to provide [...] an interesting target to search for. Interesting means that the experiment applies to a new domain or combines known and modern quantum techniques in new (potentially unexpected) ways. Subsequently, two other LLM agents judge the novelty and feasibility of the idea in the context of PyTheus. These two agents have more technical information about the PyTheus tool together with already existing implementations of setups for well-known experiments. If the idea is judged both novel and feasible, it is stored in the idea pool; otherwise, the researcher agent takes the comments of the novelty and feasibility agents into account and produces an adapted idea.

The second component of AI-MANDEL aims to implement ideas from the idea pool, written in English, in a form that PyTheus can execute. Specifically, the implementation agent receives as input (in its prompt) concrete information about the syntax of input files read by PyTheus, as well as one concrete English-language idea from the idea pool. It then translates the idea into the specific JSON syntax used by PyTheus. These JSON files encode the target of the search—for instance, the desired transformation—as well as the allowed resources and experimental techniques. Examples include the number of photons, the number of heralding detectors, and the detector types permitted in the search. Thus, the direct input to PyTheus is not natural language, but a structured JSON description of the target and the allowed experimental resources. If PyTheus cannot run the JSON script, it returns an error message to the implementation agent. Otherwise, PyTheus searches the corresponding space of optical quantum experiments and stores candidate solutions as blueprints for quantum experiments.

The candidate solutions are then verified within PyTheus by explicit simulation of the induced input-output transformation on the relevant postselected subspace and comparison with the target transformation encoded in the JSON file. Because this simulator is based on the laws of quantum optics (see the “Representation of quantum gates via colored weighted graphs” section) and does not use statistical or neural-network approximations, this verification is exact within the physical model used by PyTheus. The accessible search space is very large: It includes all experiment setups using linear optics, probabilistic photon-pair sources, deterministic photon-pair sources, and both photon-number-sensitive and nonsensitive detectors, represented in an abstract graph-theoretical form (see the “Representation of quantum gates via colored weighted graphs” section). The size of the experiments (and thus the search space) is restricted by the concrete resource constraints encoded in the JSON file produced by the implementation agent.

After PyTheus identified candidate experimental blueprints, we analyzed the resulting quantum

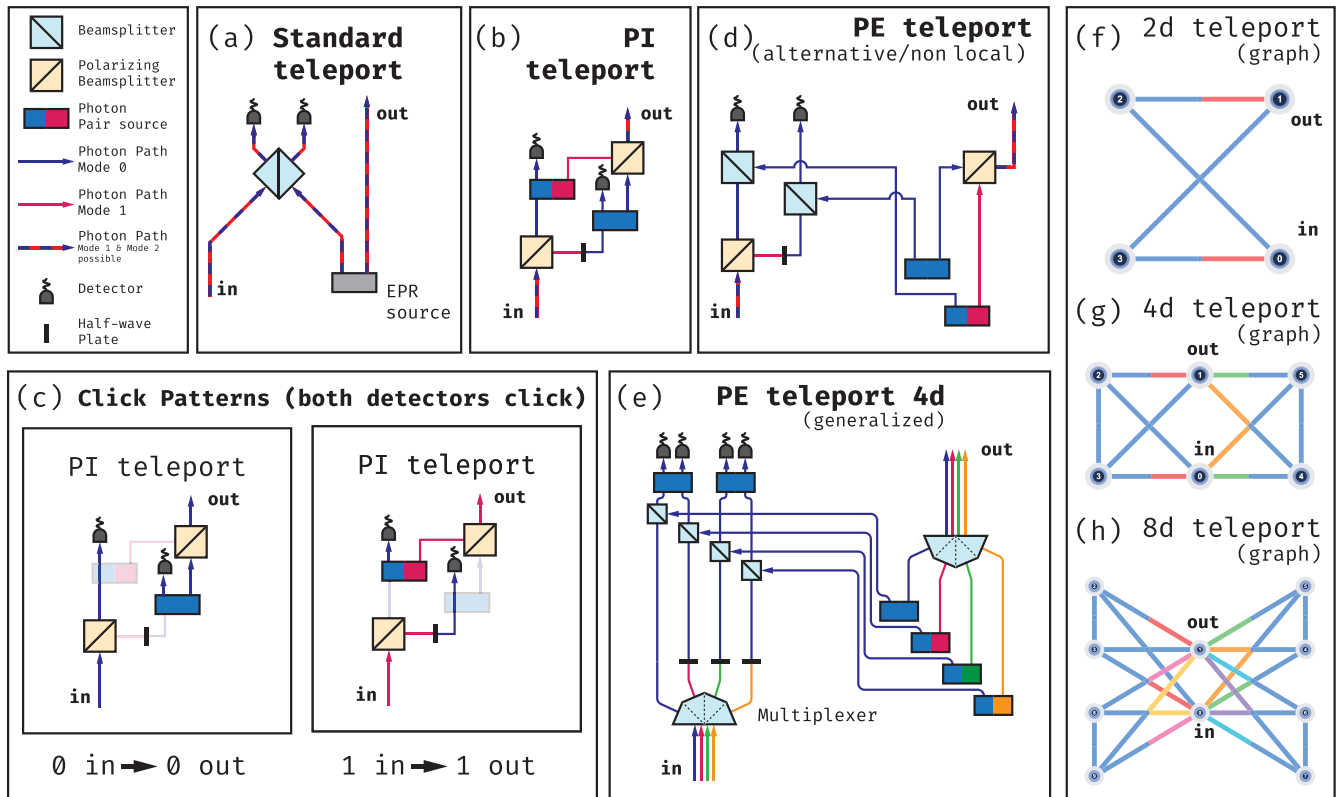


FIG. 2. Quantum teleportation as a core component of the SWAP gate. (a) The standard quantum teleportation protocol, based on a shared entangled pair and a Bell-state measurement [25,26]. (b) A variant based on path identity (PI) [20], where the quantum information of a photon is transferred to the output, while the photon itself does not propagate to the output but is instead measured. This variation was discovered accidentally during the automated search for a SWAP gate. (c) The detector click patterns corresponding to the PI analog of teleportation in panel (b). (d) A variation of the PI-based teleportation, enabled by removing the path information through quantum erasure (we label as PE teleportation). (e) A high-dimensional generalization of PE teleportation. (f)–(h) All corresponding solutions were discovered by PyTheus [27] using the graph-based representation of quantum optics [12,14]. These graphs represent the underlying logic of the quantum correlations encoded in the experiments.

transformations and interpreted the underlying physical mechanism. Human analysis therefore enters only after physically valid candidate solutions have been generated and verified within PyTheus. Our contribution at that stage is to interpret, generalize, and contextualize the mechanism. The present work reports one of the most interesting, interpretable, and generalizable of these solutions.

Representation of quantum gates via colored weighted graphs. PyTheus uses an abstract graph-based representation of quantum optics that has been largely developed in Refs. [12,14,15,31], and which is a native description of experiments based on path identity [11,13,18,20,32]. The core idea is to represent photon paths as graph vertices. Edges represent correlated photons (or single photons transitioning from an incoming to an outgoing path, which is useful for describing gates and can be understood through the Klyshko picture [33,34] or the Choi-Jamiołkowski isomorphism [35,36]). Each edge is assigned a color, corresponding to the photonic mode (for example, blue for $|0\rangle$, red for $|1\rangle$, and green for $|2\rangle$), and a weight, indicating the strength of the correlation. A joint detection event across all detectors is represented by a *perfect matching*—a subset of edges that includes every vertex exactly once. A quantum state is then described as a coherent superposition of all such perfect

matchings. For a detailed account of this formalism, we refer the reader to Ref. [27].

This graph representation is powerful because it abstracts away experimental details and focuses purely on the quantum correlation aspect of the operation, leading to readily interpretable implementations. Throughout this work, we show both graphs and concrete experimental implementations when the blueprint helps convey a critical concept. In other cases, we present only the graphs (and their perfect matchings) to describe the gate operations.

SWAP gates and teleportation. We start by looking into the SWAP gate—an operation that exchanges the quantum information of two photons,

$$\widehat{\text{SWAP}} |a, b\rangle = |b, a\rangle, \quad (1)$$

where a and b denote qubits (or qudits). The most straightforward realization is to physically send photon A to location B and photon B to location A [see Fig. 3(a)]. A conceptually more elegant approach avoids the direct transmission of the quantum carriers by using bidirectional quantum teleportation [30], i.e., teleporting $A \rightarrow B$ and $B \rightarrow A$ [see Fig. 3(b)]; each direction uses the standard teleportation protocol shown in Fig. 2(a). A teleportation-based nonlocal SWAP requires (generalized) Bell-state measurements (BSMs), motivating

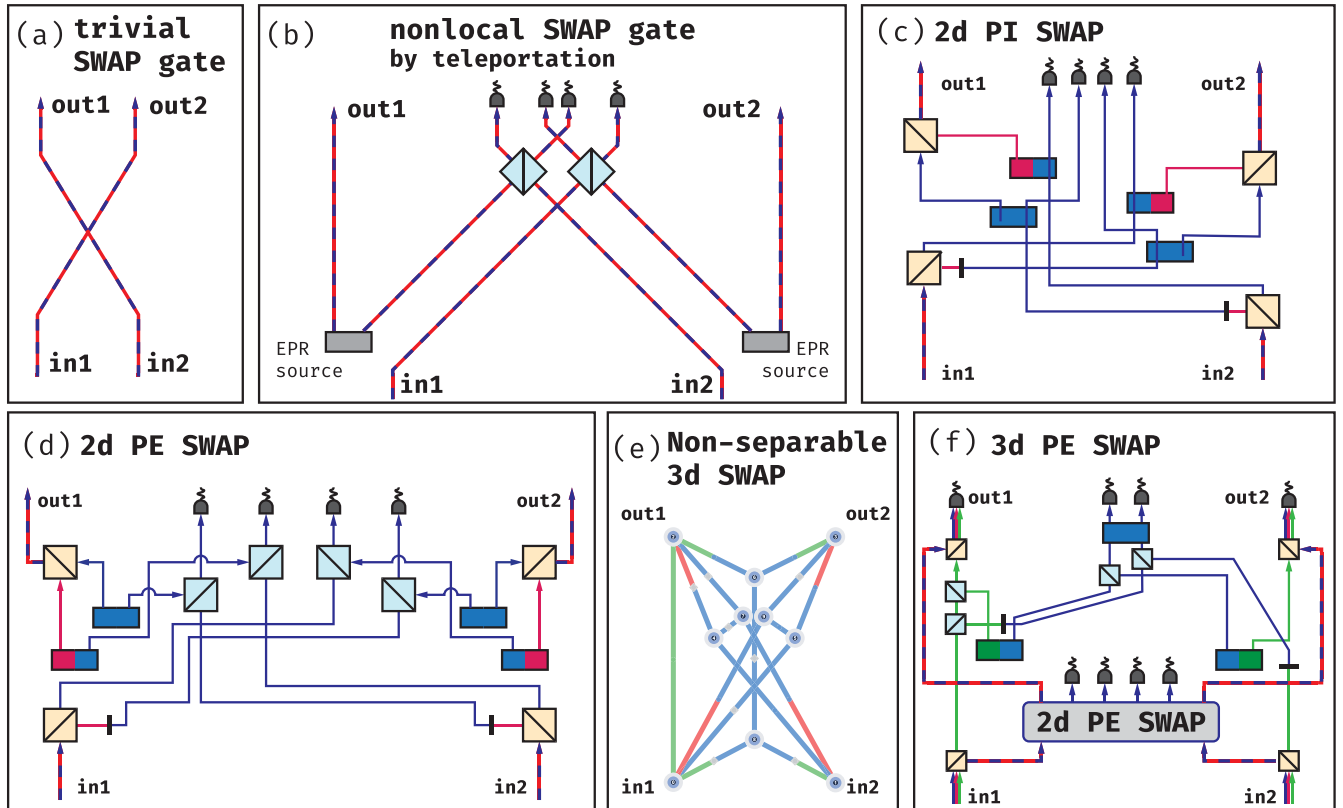


FIG. 3. Photonic SWAP gate. (a) A trivial way to perform a SWAP operation, where both photons are physically exchanged by sending each over the whole distance. (b) A SWAP gate realized through two quantum teleportations [30], without direct transmission of the photons. (c) A path identity (PI) analog of teleportation used to construct a SWAP gate. (d) An alternative solution employing path-information quantum erasure (PE). (e)—(f) A three-dimensional generalization of the SWAP gate, shown as a graph representation and as an extension of the experiment in panel (d).

solutions that boost BSM success probabilities beyond the standard linear-optics limit using ancillary resources and feed forward [37–39]. In contrast, the nonlocal photonic gates discovered here [including our PI/PE teleportation analogues in Figs. 2(b) and 2(c)] avoid both preshared entanglement and any BSM stage, instead exploiting path indistinguishability and postselection. The gates are feed forwardable at each separated location. This means that the computational photons do not need to be measured during the nonlocal operation and can subsequently be used in further local gates at each separated location.

In contrast to standard bidirectional teleportation, we discover SWAP gates that require neither Bell-state measurements nor preshared entangled photon pairs. Instead, they exploit indistinguishability through path identity as the underlying mechanism. In this process, we also find a previously unknown and generalizable pattern for constructing SWAP gates in high-dimensional quantum systems [40].

Interestingly, the generalizable pattern consists of two symmetric and spatially separated operations, in perfect analogy to the bidirectional quantum teleportation protocol. By this analogy, we have incidentally discovered a path-identity analog of high-dimensional quantum teleportation [Fig. 2(f)], for which, so far, only solutions based on generalized Bell-state measurements were known [41–43]. We refer to this way as path-identity teleportation [*PI teleportation*, Fig. 2(b)].

The PI teleportation solution operates as follows [see Fig. 2(c)]. The qubit state of the incoming photon is first split according to its polarization. For each polarization component, a photon pair can be generated such that one of the pair’s photons is path identified (i.e., overlapped) with the incoming photon. The incoming photon together with the photon in the overlapping paths is then jointly detected (at different detectors), while the remaining photon passes through a polarizing beam splitter and is directed to the remote location. Since only one photon enters the gate, we know only one of the two detectors can fire. By conditioning on detection events at both detectors, we ensure that exactly one photon-pair creation process has occurred—the specific pair being determined by the incoming photon. With this logic, the outgoing photon carries the quantum information of the incoming photon, despite never being at the same location as the incoming photon.

This mechanism is closely related to the original work on path identity by Zou, Wang, and Mandel [32,44]. The experimental implementation of teleportation based on path identity in Fig. 2(b) does not allow to spatially separate the incoming and outgoing photons due to the requirement of overlapping the paths of the photons. However, the setup can be modified: Instead of preventing the path information from ever being created (as in path-identity experiments), one can erase it at a later stage of the process. This modification introduces loss but enables genuine spatial separation in this form of

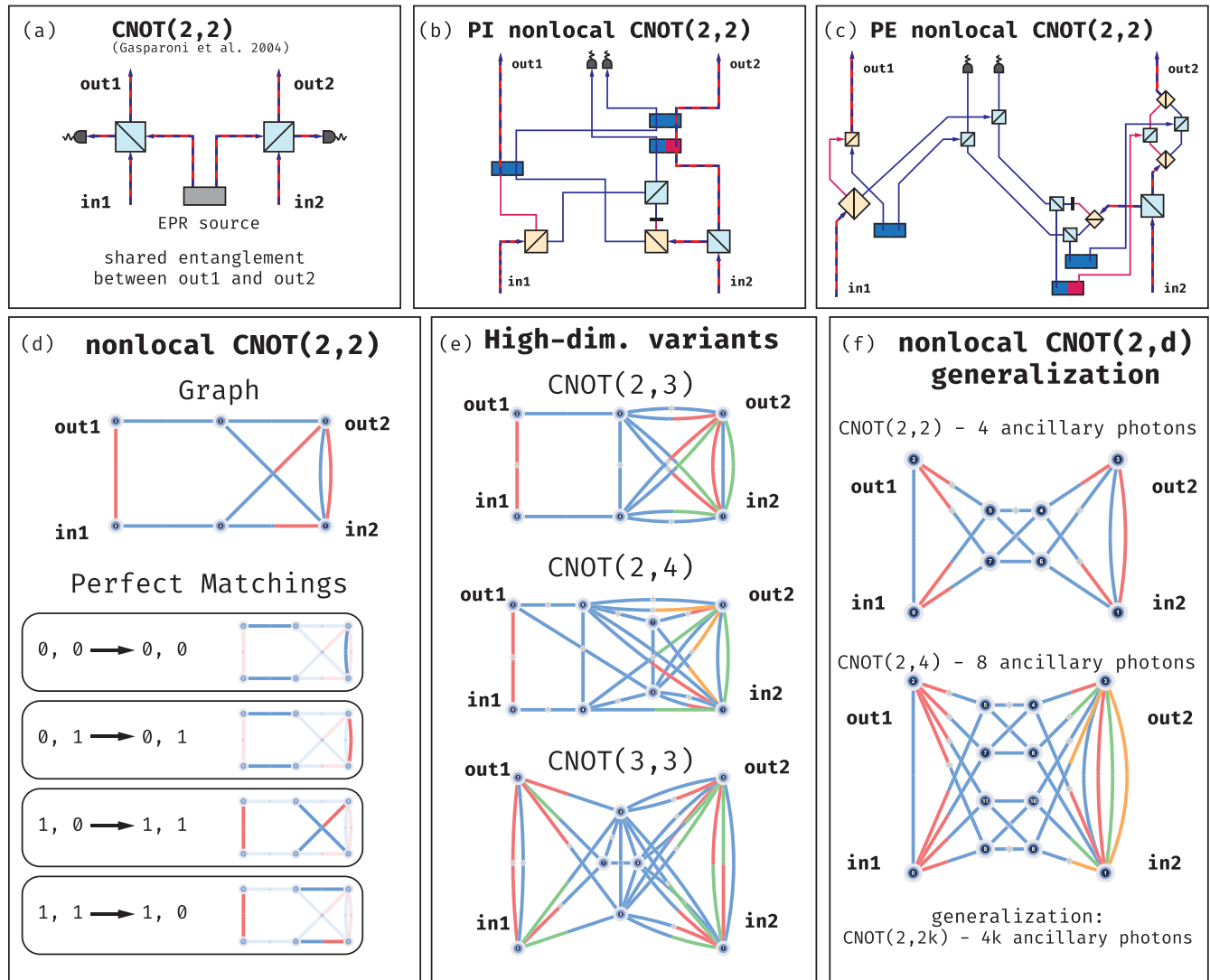


FIG. 4. Photonic CNOT gates. (a) The standard implementation of a photonic CNOT gate, in which an entangled photon pair is shared between the two sides [4]. (b) An alternative solution based on path identity that does not need shared entanglement between the input photons. (c) A new experimental blueprint for a CNOT gate realized via path-information erasure. (d) The original qubit CNOT solution discovered by PyTheus. The perfect matchings of the corresponding graph (shown below) illustrate the intrinsic working principle of the gate. (e) High-dimensional CNOT gates discovered by PyTheus. (f) Manually constructed graphs based on a generalizable pattern we identified from the machine-discovered gates. Analogous to Figs. 2 and 3, the graphs can be directly translated into implementations based on path identity or path-information erasure.

teleportationlike experiment, which we call path-information erasure teleportation [PE teleportation; for details, see Fig. 2(d)]. In a more advanced implementation, the loss introduced by these beam splitters could in principle be removed by monitoring both output ports. This, however, would require applying output-phase corrections conditioned on the corresponding detector click pattern. For higher-dimensional implementations [Figs. 2(e) and 2(f)], the solution requires more ancillary photons.

Returning to the SWAP gate, we can now use the forms of PI teleportation or PE teleportation twice (once from A to B and once from B to A) [see Figs. 3(c) and 3(d)]. For higher dimensions, we also discover implementations that do not rely on two independent processes, but instead reduce the number of ancilla photons by exploiting additional coherent superpositions of photon-pair origins in Figs. 3(e) and 3(f).

CNOT gates. Controlled-NOT operations are among the most fundamental multiphoton gates [45], where the state of one photon changes the state of the other. In the case of high-dimensional gates, it is useful to describe the gate as a controlled-X gate, where the X gate is a modulo addition [46–48],

$$\widehat{CX} |c, t\rangle = |c\rangle \hat{X}^c |t\rangle = |c\rangle |(t + c) \bmod d\rangle, \quad (2)$$

where the control c and the target t are qubit (or qudit) states, and d is the dimension of the target qudit t .

Early demonstrations of photonic CNOT gates exploited shared Bell states between the control and target photons, as in the work of Gasparoni *et al.* [4] [see Fig. 4(a)]. Since then, numerous realizations of CNOT gates assisted by entangled [49–52] or single-photon ancillas [6,7] have been reported, often relying on Bell-state measurements. Most recently, gate

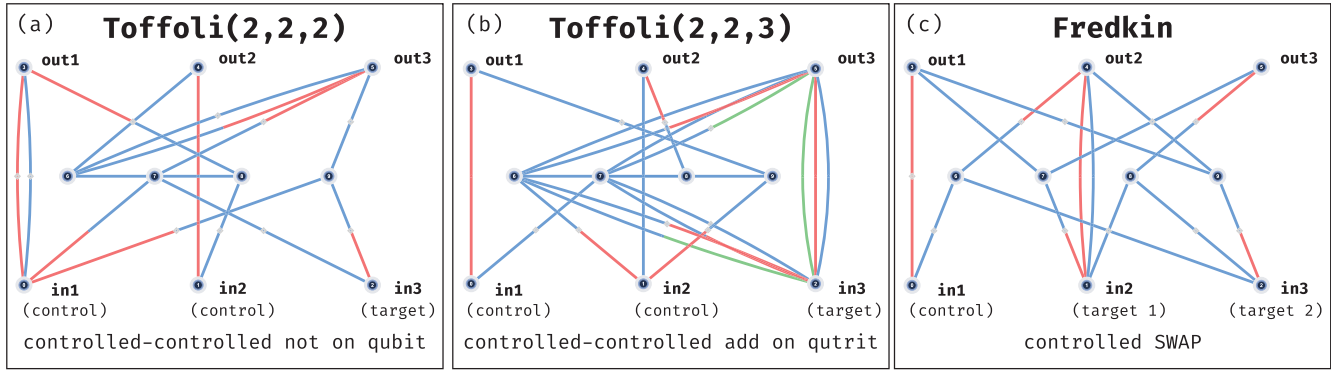


FIG. 5. Experimental proposals for three-photon gates. (a) and (b) A qubit and qutrit TOFFOLI gate, each requiring four ancilla photons. (c) A qubit FREDKIN gate implemented with four ancilla photons.

teleportation of a photonic CNOT has been demonstrated on an integrated silicon nanophotonic platform, providing an experimental realization of a teleported on-chip two-qubit logic gate [53]. Moreover, several blueprints for generalized photonic control gates in higher dimensions and with multiple photons have been proposed, encoding the desired quantum transformations into complex entangled ancillary states [5].

Using PyTheus, we have discovered solutions for CNOT gates with control space $d_c = 2$ and target space $d_t = 2, 3, 4$, as well as a genuine three-dimensional CNOT with $d_c = d_t = 3$ [see Fig. 4(e)]. A previously introduced three-dimensional CNOT required the generation of an intricate high-dimensionally entangled six-photon ancilla state [5], while our solution requires only four ancilla photons, generated from unentangled photon pairs.

The principle of the simplest, two-dimensional solution can be understood as follows. If the control photon is in state $|0\rangle$, it is directed to the ancilla detector, and an additional photon pair is generated between the second ancilla and the output. In this case, the target photon goes to its output and remains unchanged. If the control photon is in state $|1\rangle$, it is routed directly to the output; in that case, the target photon goes to the ancilla and another pair is generated with opposite states. These two options define the action of the CNOT in two dimensions [see Fig. 4(c) for the principle, Fig. 4(b) for its experimental implementation, and Fig. 4(d) for a variation]. High-dimensional generalizations use the same principles. Based on this insight, we were able to conceptualize a generalization of high-dimensional CNOT gates of arbitrary dimension in Fig. 4(f)—demonstrating that we have understood the underlying AI-discovered concept [54–56].

TOFFOLI gate. The controlled-controlled-X (TOFFOLI) gate is a fundamental three-photon gate: It advances the target only when both control photons are in the state $|1\rangle$ [58–60]. It is universal for reversible classical computation and forms the basis of arithmetic and oracle constructions in many quantum algorithms. Its operation is defined as

$$\begin{aligned} \widehat{CCX} |c_1, c_2, t\rangle &= |c_1, c_2\rangle \hat{X}^{(c_1 \wedge c_2)} |t\rangle \\ &= |c_1, c_2\rangle |t + (c_1 \wedge c_2) \bmod d\rangle. \end{aligned} \quad (3)$$

Here, $c_1, c_2 \in 0, 1$ are the control qubits, t is the target qubit (or qudit), d is the dimension of the target system, and $\hat{X}$

denotes the generalized Pauli-X (modulo-addition) operator. The symbol $\wedge$ represents the logical AND, which evaluates to 1 if and only if both controls are 1, and 0 otherwise.

PyTheus discovers solutions for the TOFFOLI gate for $c_1, c_2 \in \{0, 1\}$ and $t \in \{0, 1\}$ (which is the standard qubit case) and for $t \in \{0, 1, 2\}$, which is a generalized high-dimensional case. The solutions are displayed in Figs. 5(a) and 5(b). All solutions share the same properties: The three photons never meet at a single location, and there are no entangled photon pairs shared between the photons. Rather, the shared resource is the indistinguishability of the origins of the photon pair's origin. Surprisingly, despite these critical resource constraints, it is possible to achieve these goals with just four ancillas.

FREDKIN gate. The FREDKIN gate (also known as controlled-SWAP or CSWAP) is another fundamental three-photon gate: It swaps the two target qubits only when the control is in state $|1\rangle$ [59,61,62]. It is important for the SWAP test used in overlap and fidelity estimation [63,64] and for data-dependent routing of quantum information without measurement [65,66]. Its operation is defined as

$$\begin{aligned} \widehat{CSWAP} |c, a, b\rangle &= |c\rangle \widehat{SWAP}^c |a, b\rangle \\ &= \begin{cases} |0, a, b\rangle, & c = 0, \\ |1, b, a\rangle, & c = 1. \end{cases} \end{aligned} \quad (4)$$

Here, $c, a, b \in \{0, 1\}$ are qubit states (we restrict to $d = 2$), and $\widehat{CSWAP}$ is the controlled-SWAP operator, where $\widehat{SWAP}$ (or $\hat{S}$) is defined by its action $\hat{S} |x, y\rangle = |y, x\rangle$ on computational-basis states.

Using PyTheus, we discover an experimental implementation for the FREDKIN gate that requires only four ancillas. The solution is shown in Fig. 5(c). As before, none of the photons meet at a single location (i.e., they do not interact) and no entangled pairs are shared among the photons.

Error estimation. We consider two dominating sources of error in experiments based on linear optics and SPDC sources: (1) the creation of additional photons due to the probabilistic nature of photon-pair generation and (2) photons entering a detector but not being registered because the detector efficiency is below 100%. We calculate how these two factors affect the heralding probability and the fidelity of the output states.

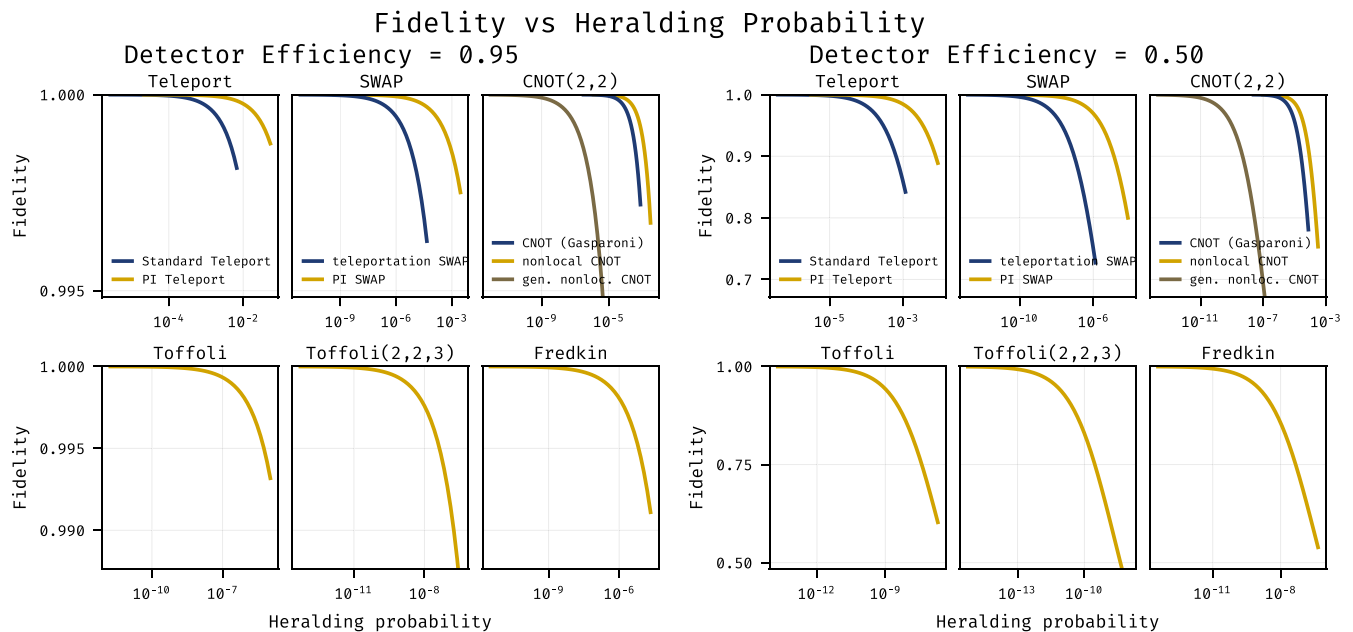


FIG. 6. Error estimation. For the setups shown in Figs. 2–5, we plot the output-state fidelity F vs the heralding probability P_{her} under the error model described in the “Error estimation” section. The fidelity here is defined as the average over the fidelities of all output states, for all basis states allowed at the input. The heralding probability is the probability of producing the target click pattern. The source amplitude g is varied from 0.01 to 0.5. Left: detector efficiency $\eta = 0.95$. Right: detector efficiency $\eta = 0.50$. Where applicable, we compare the proposed schemes with previously known implementations, including standard teleportation and the Gasparoni CNOT gate, evaluated under the same source and detector model. The code for computing the quantities plotted here can be found on GitHub [57].

(1) *Higher-order terms*—A probabilistic photon-pair source can be modeled by $|\Psi\rangle \approx (1 + g a^\dagger b^\dagger + \frac{g^2}{2} (a^\dagger b^\dagger)^2 + \mathcal{O}(g^3))|\text{vac}\rangle$. Taking the pump amplitude g to be a global factor for all nonlinear crystals in a setup, the probability of creating N photon pairs simultaneously is $\sim |g^N|^2$. During the optimization, we ignore events in which more photons are created than leave the setup (to ancillary detectors and outgoing paths). This approximation is justified because for low pump amplitudes ($|g| \ll 1$), higher-order terms become increasingly unlikely. In principle, such higher-order events can be identified and excluded using photon-number-resolving (PNR) detectors. However, if the PNR detectors are imperfect, these events cannot always be rejected.

(2) *Detector inefficiency*—Our setups rely on the coincidence detections from multiple PNR detectors firing at the same time. The resulting quantum state is a superposition of all possible paths the photons could have taken to arrive at the detectors. If one detector fails to register a photon, the corresponding event does not produce the correct click pattern and is therefore rejected during postselection. High-end detectors can reach efficiencies of 0.95, whereas more basic detectors may have an efficiency of 0.5.

The combination of higher-order terms and detector inefficiency can lead to the heralding of events that are not part of the target process. For example, if one photon is detected in each detector and each outgoing path, this pattern may in fact result from the creation of two excess photons, followed by photon loss at the detectors. Experimentally, we cannot distinguish such event from the desired ones considered during the optimization. As a result, the output state has a fidelity below 1. For sufficiently low values of g , the fidelity remains

close to 1, since higher-order terms are very rare in this regime. However, a low pump amplitude also decreases the heralding probability, leading to a very low rate of successful gate operations. In Fig. 6, we show the *fidelity versus heralding probability* curves as g is varied from 0.01 to 0.5 for the setups presented in this paper.

From these estimates, it is clear that the new schemes presented here are at least as robust under these two error sources as the known results. It will be interesting to see whether a real-world implementation that has numerous other noise sources (dark counts, imperfect indistinguishability, source instabilities, and others) can match or improve upon existing realizations.

Outlook. We have shown that the idea of indistinguishability by path identity is not only a resource for quantum interference and state generation but also a powerful quantum for the design of new forms of quantum gates. Accidentally, while conceptualizing the solutions, we discover an analog to quantum teleportation—which is by itself an interesting finding. The presented experimental blueprints only use previously demonstrated technologies, which indicates that they are also experimentally feasible. An interesting future question is how path identity and the solutions presented in this paper can be used for making quantum networks more efficient.

The unique way in which indistinguishability is generated via path identity—not by erasing the information *post hoc*, but by never letting it be generated—might enable new experiments in causally indefinite processes, e.g. for superpositions of orders of quantum gates [67–70].

Another interesting question is whether other forms of indistinguishability, such as that of independent single

photons, can play a role analogous to path identity in non-local gate design. While path identity in our schemes relies on coherent uncertainty about the origin of photon pairs, single-photon implementations would instead exploit particle indistinguishability between independently generated photons. Recent progress in the use of indistinguishable single photons for the generation of intricate quantum states suggests that this may be a promising direction [31,71–73].

The work presented here not only points to new directions in quantum optics but also highlights a concrete instance of AI-driven scientific workflow. The core experimental blueprints were automatically discovered with PyTheus, which searches large quantum-optics design spaces for setups satisfying specified targets. The overall research direction and initial question for this paper were generated by AI-MANDEL [24], an LLM-agent system that proposes and filters hypotheses and can translate selected ideas into tool-compatible discovery tasks—specifically can automatically use PyTheus. Looking ahead, further advances in such agentic automation could enable closed-loop pipelines

that iterate between hypothesis generation, constraint-aware design, robustness checks, and experimental considerations, thereby expanding both the scale and the systematicity with which new photonic primitives are explored. Beyond the present project, an additional extension is automated generalization via metadesign [74], which could elevate individual discovered instances into broader families and design principles.

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Data availability. There are no publicly available research data or software supporting this manuscript. Requests for further information or data should be sent to the authors.

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